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To the Editorial Office / Editor-in-Chief
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Dear Editor,

We are pleased to submit our manuscript, “**The Fidelity–Utility Paradox in Surrogate-Based Reinforcement Learning: A Role-Separated Hybrid Methodology in an HVAC Control Case Study**,” for consideration as an original research Article in *AI*.

Deep reinforcement learning (DRL) for building HVAC control almost universally relies on fast learned surrogates as training environments, under the tacit assumption that a more predictively accurate surrogate is a better environment in which to train a controller. To our knowledge, our study is the first to test this assumption directly and, under a pre-specified, version-locked protocol, to report that it can fail. On the BOPTEST `bestest_air` benchmark, a calibrated grey-box physical twin that is more than twice as accurate as a coarse black-box surrogate (24-hour rollout RMSE 0.644 vs. 1.557 °C) nevertheless yields controllers that collapse on every seed on the live simulator, whereas the weaker surrogate trains a usable controller. We show that the operative variable is the surrogate’s temporal resolution (its coarse per-step increment) rather than model class or predictive accuracy, and we resolve the paradox with a role-separating hybrid: black-box rollouts combined with a frozen physics twin used only as a per-step model-disagreement reward censor, which recovers robust control at roughly 85× live-simulator throughput. A pre-specified transferability study then delineates a precise, component-level boundary: the inverse-calibration pipeline transfers across a hydronic family of buildings, whereas a frozen policy does not.

We believe this work is a strong fit for *AI* because it speaks directly to how learned world models and surrogate environments should be evaluated and used for control. HVAC serves as the concrete case study for a broader surrogate-based-RL hypothesis: rather than proposing yet another controller, the paper provides a corrected, quantified account of *why* surrogate-trained controllers succeed or fail, together with a reusable role-separating recipe, of practical significance for the growing literature on physics-informed digital twins and safe RL for buildings.

This manuscript is original, has not been published previously, and is not under consideration elsewhere. All authors have read and approved the submitted version and agree to its submission to *AI*. The authors declare no conflicts of interest, and this simulation-only study involves no human or animal subjects. This research was funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant No. AP23488794). Thank you for considering our submission; we look forward to the reviewers’ comments.

Sincerely,

Almaz Sapargali
on behalf of all co-authors
(Aksultan Mukhanbet, Serik Aibagarov, Beimbet Daribayev, Shona Shinassylov, Paulo Trigo)